

An Introduction to Measure Theory - Chapter 1

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1 Exercise 1.1.1 - Boolean Closure

Show that if $E, F \subset \mathbb{R}^d$ are elementary sets, then the union $E \cup F$, the intersection $E \cap F$, and the set theoretic difference $E \setminus F := \{x \in E : x \notin F\}$, and the symmetric difference $E \Delta F := (E \setminus F) \cup (F \setminus E)$ are also elementary. If $x \in \mathbb{R}^d$, show that the translate $E + x := \{y + x : y \in E\}$ is also an elementary set.

An **elementary set** is that which is a union of a finite amount of boxes. Thus, let $\bigcup_i E_i$ and $\bigcup_j F_j$ be the unions of the finite number of boxes comprising sets E and F , respectively.

$$E \cup F = \bigcup_i E_i \cup \bigcup_j F_j = \bigcup_i \bigcup_j E_i \cup F_j \quad (1)$$

$$E \cap F = \bigcup_i E_i \cap \bigcup_j F_j = \bigcup_i \bigcup_j E_i \cap F_j \quad (2)$$

$$E \setminus F = \bigcup_i E_i \setminus \bigcup_j F_j = \bigcup_i \left(E_i \cap \neg \bigcup_j F_j \right) = \bigcup_i \left(E_i \setminus \bigcup_j F_j \right) \quad (3)$$

This one tripped me up a bit because elementary sets aren't closed under set negation, and I was inherently proving the closure of elementary sets under the set difference for some reason.

$$E \Delta F = E \setminus F \cup F \setminus E = \bigcup_i \left(E_i \setminus \bigcup_k F_k \right) \cup \bigcup_j \left(F_j \setminus \bigcup_k E_k \right) = \bigcup_i \bigcup_j \left(E_i \setminus \bigcup_k F_k \cup F_j \setminus \bigcup_k E_k \right) \quad (4)$$

$$E + x = \bigcup_i (E_i + x) \quad (5)$$

The set theoretical operations of union, intersection, difference, and symmetric difference, and translation, between two elementary sets yield a union of finitely countable boxes, which is itself an elementary set. $\square$